If you though that the CPU was the heart of your computer, then think again...

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The other day my friend's SMPS gave out, and guess who got the call to look for a new one? Moi! So we went to Lamington Road in Mumbai. It wasn't really necessary since many e-commerce web sites now sell most brands and will ship it to your doorstep for a small price, however, we were bored and our next trip to Lamington was due, so we thought "why not?"

SMPS are one of the most neglected components of your computer but of late have gained importance mainly due to power-hungry GPUs. Before getting a new PSU for a new build, the first thing you need to do is calculate the net wattage. You can use any of the many PSU calculators available on the internet to get the job done. A particularly good one is eXtreme Outer Vision's PSU calculator. Select the components that you have or will be having in your system to get the net wattage required for your SMPS.

Once that's taken care of, start counting the number of connectors you'll need. High-end motherboards these days have upto 2x 8-pin ATX connectors for the extreme edition processors. Then there's the 24-pin connector for your motherboard which will be present in all PSUs. All your HDDs, ODDs and SSDs will use the SATA power connectors, hence, the more the merrier. Floppy connectors are no longer needed and molex connectors are needed for fans and, on certain motherboards, to offset the load from the main 24-pin connector. There's the gauge of the wire too; this is important as current passing through a thin wire will likely melt it and lead to unnecessary wastage of time in RMA'ing the device. The most important of all is the PCI-E which will be a 4/6/8-pin connector that goes into your GPU. Almost all PSUs come with adapters which utilise 2x 4-pin molex connectors and have a PCI-E connector on the other end.

Most PSUs have a multi-rail design, which means that the amperage is divided into multiple sources. So you can have limits on particular rails. The downside of this is that often one main rail needs to be loaded (needs something drawing power from it) in order for other rails to activate. Also, it's a bit costlier to have a multi-rail design and manufacturers often falsely claim that their devices have a multi-rail design when they actually have a single-rail design.

Platinum PSUs have an efficiency of >90%

Having a modular PSU helps in cable management as unwanted connectors can be detached when not required. Other than that, there's no major reason for going in for a modular PSU.

We've seen those badges on PSUs that say 80+ Bronze/Silver/Gold/Platinum. These are certifications that the particular model has received from an independent authority called Ecova Plug Load Solutions. Cheap PSUs supplied along with cabinets tend to have an efficiency of 65-70 per cent while mainstream brands have 85 per cent efficiency (these are 80+ Bronze PSUs available at just the entry-level). To see the difference it makes, let's consider an output of 400 watts. The cheap PSU will draw (400/.70)=570W and an 80+ Bronze PSU will draw (400/.85)=470W. The 80+ Bronze PSU draws a whopping 100W less than the cheaper PSU. In the long run, an efficient power supply pays for itself several times over.

What came as a surprise was that most companies reduced the warranty period on their products a few months back – five years became three and three became one.

There are plenty of brands to choose from, and a little known fact is that most brands don't manufacture their own PSUs. So going in for the OEM that actually manufactures the PSU can save you lots of cash. Seasonic, Channel Well Technology and FSP are such OEM manufacturers whose models are widely used by others. Also these brands have their efficiency rated at 50°C, which is a realistic scenario compared to PSU which are rated at 25°C.

With all that in mind, we settled for a Seasonic SS-750JS which is a pretty decent 750W PSU at a great price, and fitted our needs beautifully.